3.4.2 When electronic sensing isn’t enough

The limitations I encounter in building my home digital twin mirror challenges found in industrial deployments at every scale. Whether you’re monitoring a single home or a sprawling manufacturing facility, you’ll face the same constraints, including that some measurements are too expensive to fully automate, some equipment lacks sensor interfaces, some data is locked in proprietary systems, and some parameters require human observation or interpretation. The difference lies not in the nature of these challenges but in their magnitude—an industrial plant might have thousands of analog gauges instead of one, or deal with vendor lock-in across dozens of equipment suppliers rather than a single utility company. Understanding these constraints through simple home examples helps you recognize and plan for similar situations in more complex environments:

- *Cost constraints*—a comprehensive electrical monitoring system with current transformers on every circuit would cost thousands of dollars. Rather than make this substantial investment, I monitor whole-home consumption with a single optical sensor on my electricity meter, supplemented by smart plugs or current transformers on major appliances.
- *Technology gaps*—I am unable to attach a sensor to my solar inverter to measure electricity production. The inverter does host a simple API available over my local home network that I can query for production data.
- *Existing infrastructure*—my smart electricity meter collects detailed consumption data every 30 minutes, but my utility only provides daily summaries through their web portal. The detailed data exists, but isn’t accessible through any API.
- *Complex measurements*—pool water chemistry requires measuring pH, chlorine levels, alkalinity, and other parameters. Automated sensors for all these measurements would cost more than my entire digital twin project.